

Design and Implementation of a Modern Data Warehouse using Medallion Architecture

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Institution/Company: Portfolio Project

Project GitHub Repository: [Link](#)

Abstract

This thesis-style report presents the end-to-end design and implementation of a modern, enterprise-grade data-warehouse solution that consolidates fragmented ERP and CRM data by adopting the Medallion Architecture (Bronze → Silver → Gold). The project demonstrates deterministic bulk-load ingestion, rigorous cleansing and enrichment via stored procedures, star-schema dimensional modelling, and a repeatable data-quality framework. All artefacts—SQL DDL, ETL code, quality-check suites, and metadata catalogues—are drawn from the accompanying public GitHub repository, ensuring complete reproducibility.

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1 Introduction

Large organisations typically store operational data in disparate source systems. Without a centralised integration layer, analysts confront conflicting keys, inconsistent semantics, and missing audit trails. This project resolves those deficiencies by building a single-source-of-truth data platform in SQL Server, underpinned by proven dimensional-modelling theory (Kimball & Ross, 2013).

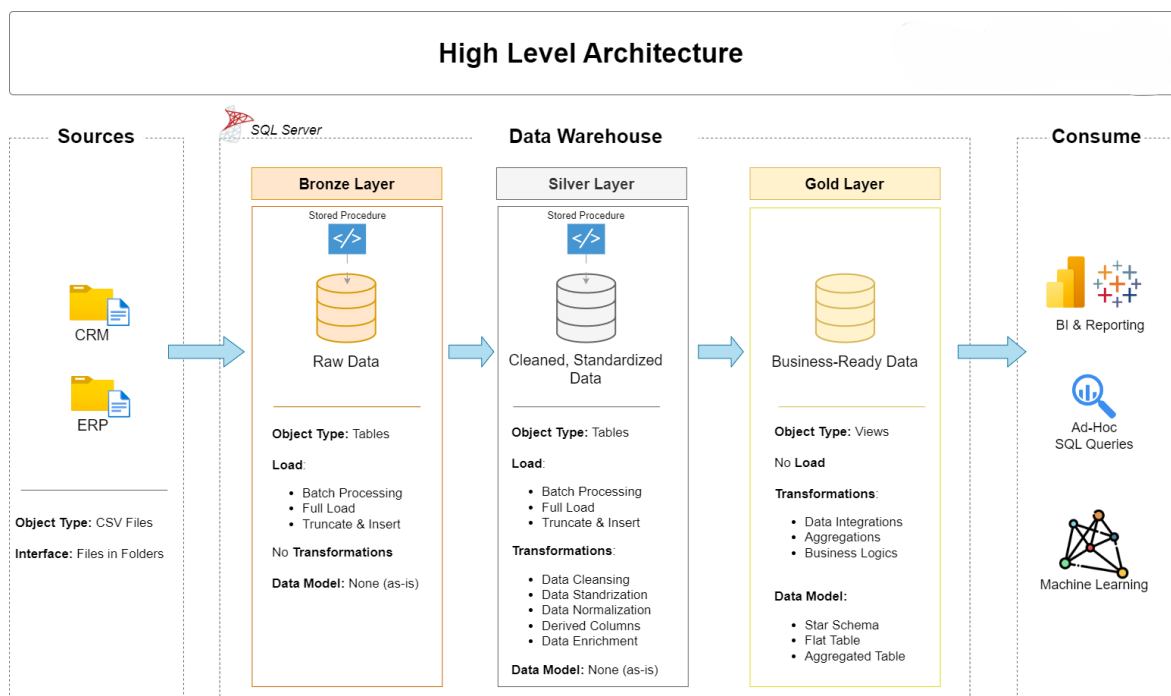
2 Business Objectives & Requirements

- ✓ Integrate ERP and CRM feeds into a single database in less than 2 hours nightly.
- ✓ Guarantee referential integrity across customer, product and sales domains.
- ✓ Expose a conformed star schema to downstream BI tools (Power BI, Tableau).
- ✓ Document every table, column and transformation step for self-service analytics.
- ✓ Embed automated data-quality tests that fail the build when critical rules are violated.

3 Architectural Overview

The design follows the Medallion pattern:

- Bronze – raw, immutable copies of every source table.
- Silver – cleaned and standardised records ready for joins.
- Gold – business-oriented star-schema views optimised for analytics.



4 Technology & Methodology

Category	Tool / Standard	Role in Solution
RDBMS	SQL Server 2022	Storage engine, T-SQL runtime
Scripting	T-SQL Stored Procedures	ETL orchestration
File Interface	CSV via BULK INSERT	Source-system hand-off
Version Control	Git	Change management, peer review
Diagramming	Draw.io	Architecture & lineage artefacts
Modelling Approach	Kimball Star Schema	Analytic query performance
Layering Pattern	Medallion (Bronze / Silver / Gold)	Isolation & lineage
QA Method	SQL QC scripts + assertions	Data-integrity gate
DevOps / Scheduling	SQL Agent (proposed)	Automation & monitoring

5 Database Initialisation & Schema Strategy

Script `init\_database.sql` enforces idempotent deployment by dropping and recreating the `DataWarehouse` database, then creating three dedicated schemas:

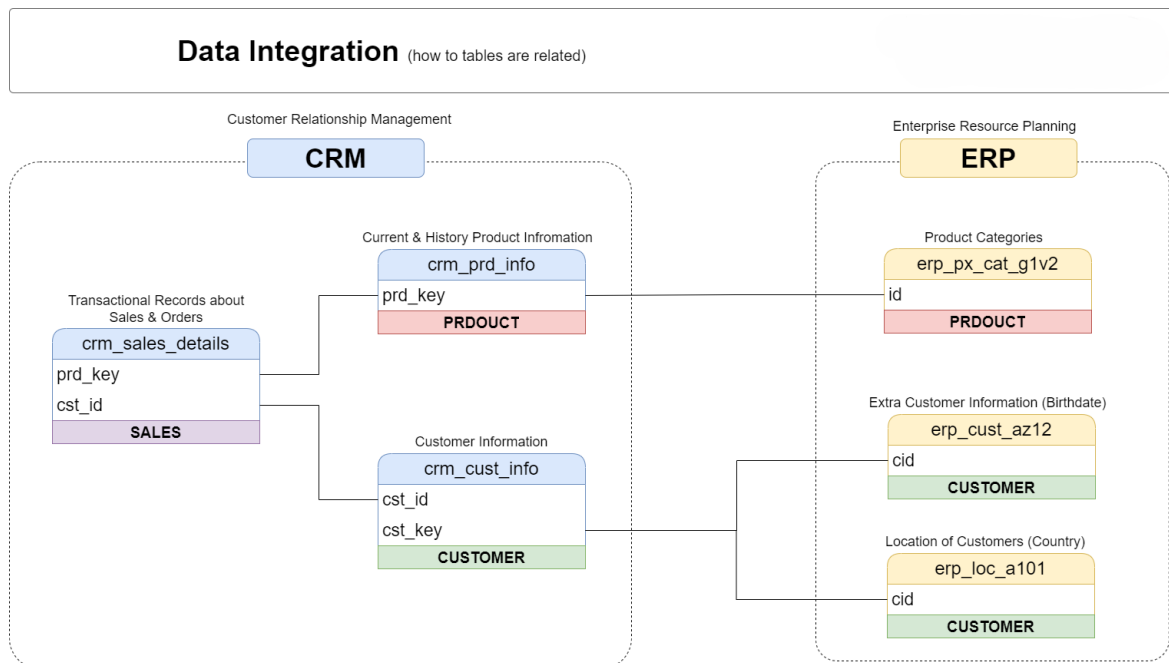
```
CREATE DATABASE DataWarehouse;  
GO  
CREATE SCHEMA bronze;  
CREATE SCHEMA silver;  
CREATE SCHEMA gold;
```

Layer isolation enables explicit ACL policies (e.g. analysts: SELECT on gold only) and prevents cross-layer contamination.

6 Bronze Layer – Raw Ingestion

The stored procedure `bronze.load\_bronze` bulk-loads six CSV extracts (three CRM, three ERP). Each target table is truncated to maintain idempotence. Example extract:

```
BULK INSERT bronze.crm_cust_info  
FROM 'C:\sql\dwh_project\datasets\source_crm\cust_info.csv'  
WITH (FIRSTROW = 2, FIELDTERMINATOR = ',', TABLOCK);
```



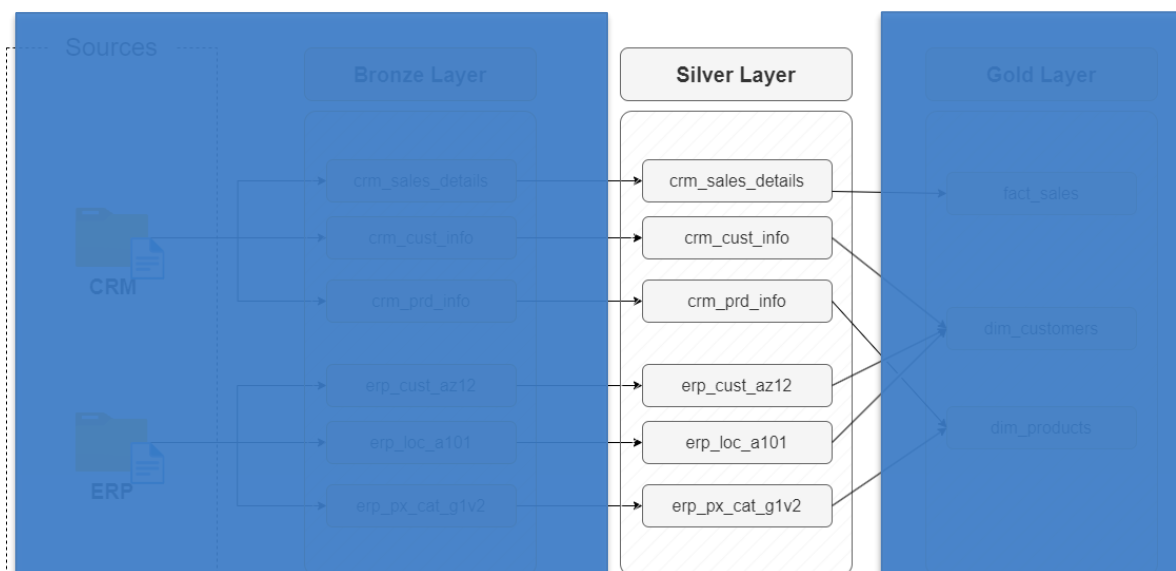
7 Silver Layer – Cleansing & Enrichment

`silver.load\_silver` transforms Bronze data by:

- Converting single-character codes to descriptive enums (e.g. gender 'M'→'Male').
- Removing duplicates using window-function row-numbering.
- Generating slowly-changing product end-dates via `LEAD()`.

```
CASE WHEN UPPER(TRIM(cst_gndr)) = 'F' THEN 'Female'  
      WHEN UPPER(TRIM(cst_gndr)) = 'M' THEN 'Male'  
      ELSE 'n/a' END AS cst_gndr
```

Data Flow (data Lineage)

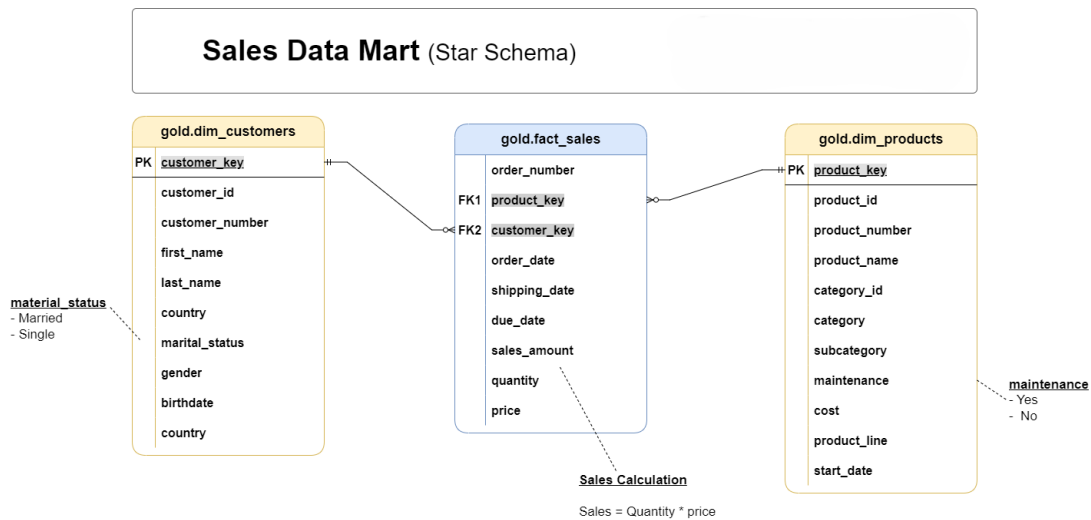


8 Gold Layer – Dimensional Modelling

Gold views construct a canonical star schema:

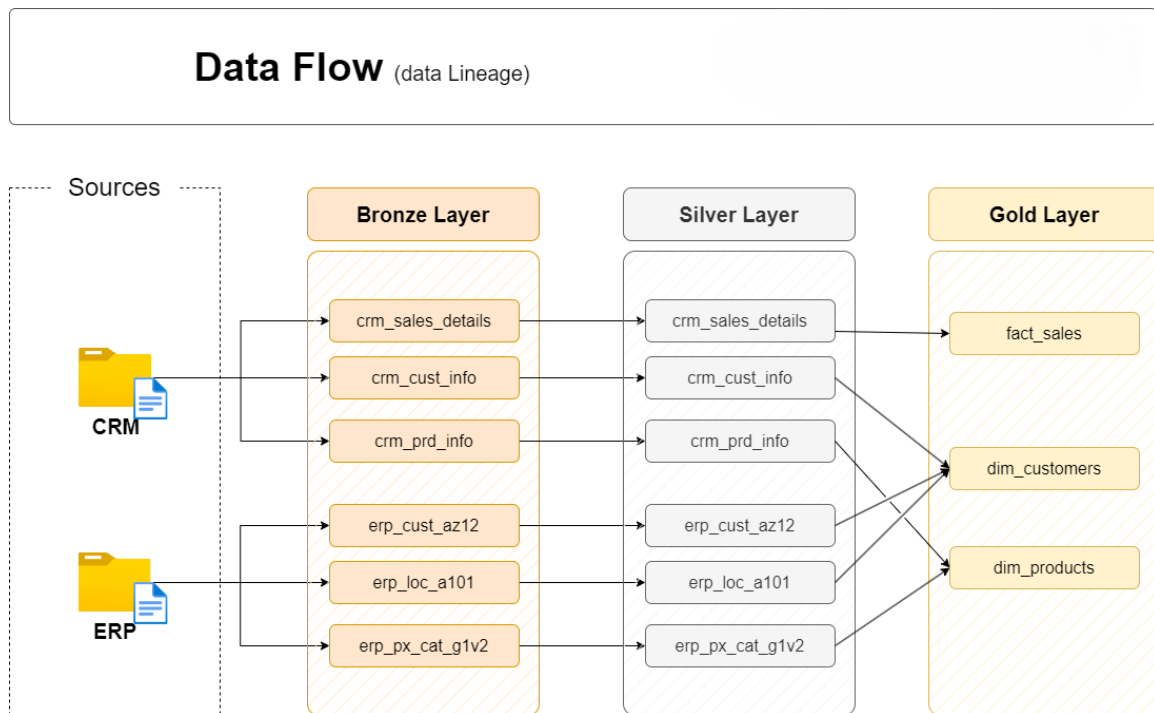
- `gold.dim\_customers` – customer attributes with surrogate `customer\_key`.
- `gold.dim\_products` – enriched product hierarchy data.
- `gold.fact\_sales` – additive measures and foreign keys.

```
SELECT ROW_NUMBER() OVER (ORDER BY cst_id) AS customer_key,  
       ci.cst_id      AS customer_id,  
       ci.cst_key     AS customer_number,  
       ...  
FROM silver.crm_cust_info ci  
LEFT JOIN silver.erp_cust_az12 ca ON ci.cst_key = ca.cid;
```



9 End-to-End ETL Pipeline

The nightly pipeline orchestrates `load\_bronze` ► `load\_silver` ► materialise Gold views. Execution metadata (start/end timestamps) is logged in each procedure for auditability.



10 Data-Quality Assurance Framework

Quality scripts test 98 distinct rules. Critical examples:

- Duplicate surrogate keys in dimensions – expect 0 rows.
- `fact\_sales` foreign-key lookups – expect no orphan rows.
- `Sales = Quantity × Price` invariant.

```
SELECT *  
FROM gold.fact_sales f  
LEFT JOIN gold.dim_products p ON p.product_key = f.product_key  
WHERE p.product_key IS NULL;
```

11 Metadata & Data-Catalogue Management

The file `docs/data\_catalog.md` documents every Gold table. An excerpt for `gold.dim\_customers` is reproduced below.

Column	Type	Description
customer_key	INT	Surrogate key
customer_id	INT	Natural CRM identifier
country	NVARCHAR(50)	Country of residence

12 Naming-Convention Governance

Strict patterns (e.g. `dim\_`, `fact\_`, suffix `\_key`) minimise ambiguity and accelerate developer onboarding. All stored procedures follow `load\_<layer>`.

13 Analytical & Reporting Capabilities

The Gold schema powers:

- Monthly sales trend dashboards.
- Customer segmentation (age, gender, geography).
- Product profitability heat-maps.

14 Results, Lessons Learned & Future Work

Deliverables include a reproducible database, automated ETL, full QC suite, and complete documentation. Future enhancements: change-data-capture incremental loads, SCD Type-2 history, and row-level security.

15 Appendix

A. Key SQL scripts (see GitHub repository).

B. Sample source CSV records.

16 References

Kimball, R. & Ross, M. (2013). The Data Warehouse Toolkit (3rd ed.).

Microsoft Docs. SQL Server 2022 Online Documentation.

Project GitHub Repository: <https://github.com/KishanPatel25/Industry-Standard-SQL-Data-Warehouse-Project>